

Learn to use For loop with p5 JS library

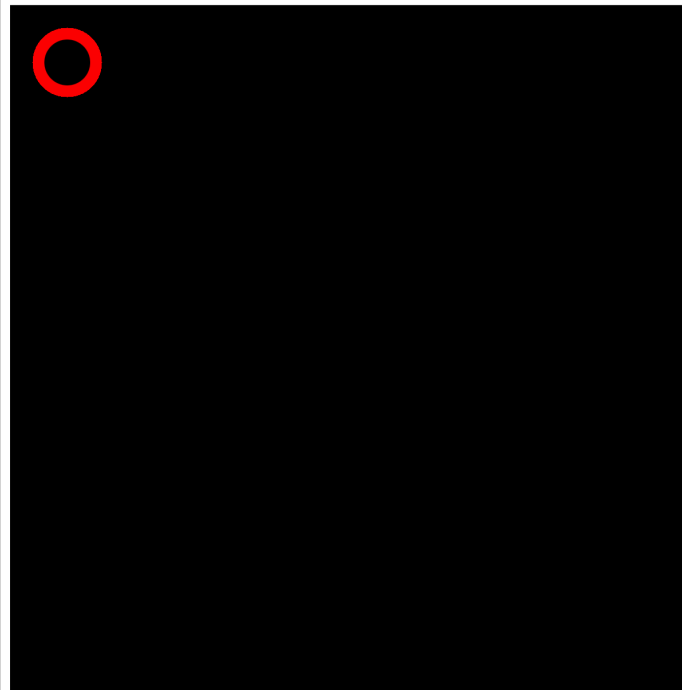
Problem Statement:

Draw 10 circles diagonally across the canvas (size 600 by 600) using gradient colours.

Step by Step:

1. Start with a for loop to draw one circle

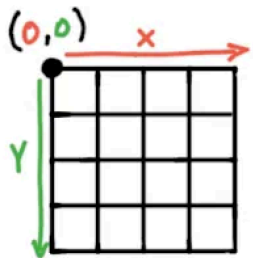
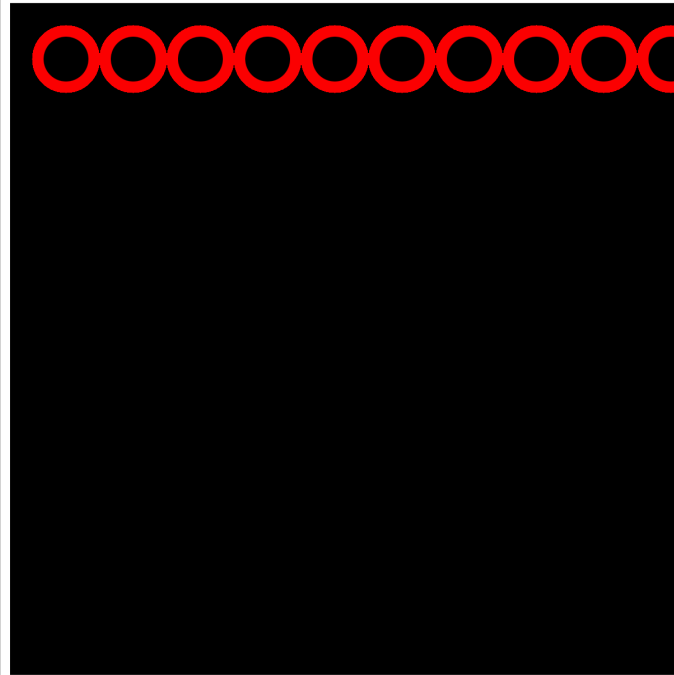
```
1. // run once
2. function setup() {
3.   createCanvas(600, 600);
4.   background("black");
5. }
6.
7. // run 60 times per one second
8. function draw() {
9.   for (let count=0; count<10; count++) {
10.    fill("black");
11.    stroke("red");
12.    strokeWeight(10);
13.
14.    circle(50,50,50);
15.   }
16. }
```



If you cannot see this, use JS console to check for any spelling error or missing { }.

2. Use a xpos variable

```
8. // run 60 times per one second
9. function draw() {
10.   let xpos = 50;
11.
12.   for (let count=0; count<10; count++) {
13.     fill("black");
14.     stroke("red");
15.     strokeWeight(10);
16.
17.     circle(xpos, 50, 50);
18.     xpos = xpos + 60;
19.   }
20. }
21.
```

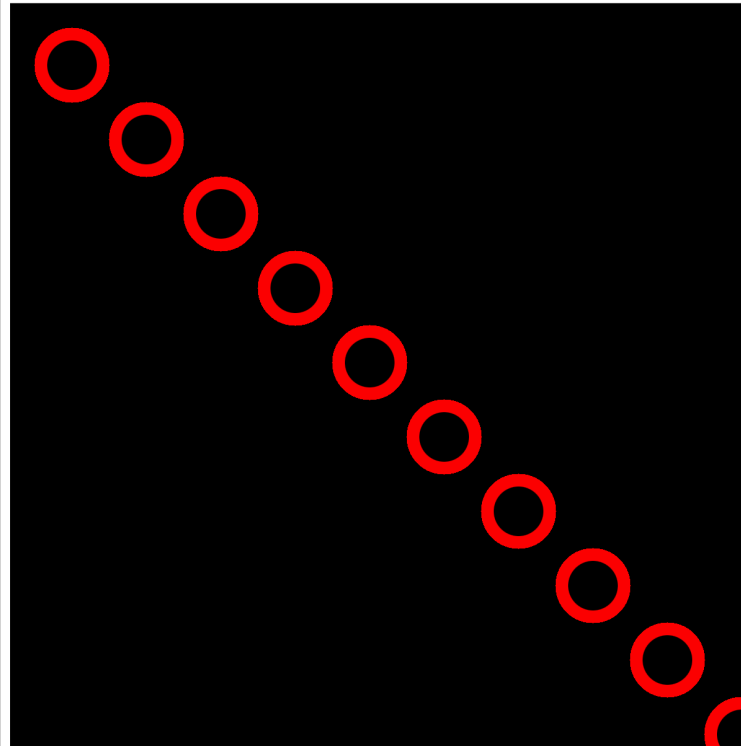


2D

TIP:
Remember p5 JS coordinate system starts
from top-left corner, with X=0, Y=0

3. Let's add in a ypos variable

```
8. // run 60 times per one second
9. function draw() {
10.   let xpos = 50;
11.   let ypos = 50;
12.
13.   for (let count=0; count<10; count++) {
14.     fill("black");
15.     stroke("red");
16.     strokeWidth(10);
17.
18.     circle(xpos, ypos, 50);
19.     xpos = xpos + 60;
20.     ypos = ypos + 60;
21.   }
22. }
23.
```



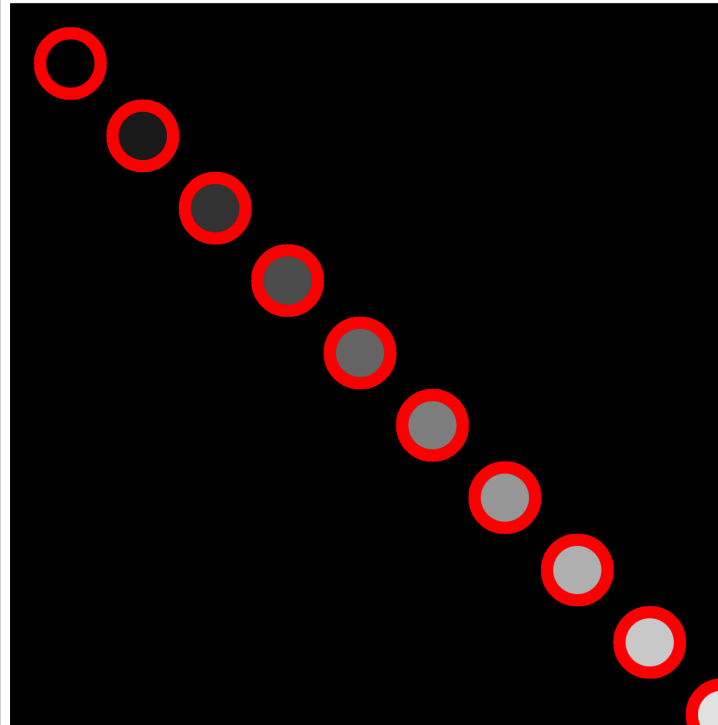
TIP:

```
// fill shape with colour
fill( "red" ); // colour name is okay
fill( 220 ); // gray colour using number is okay too
```



4. Let's do the same for colour gradient.

```
8. // run 60 times per one second
9. function draw() {
10.   let xpos = 50;
11.   let ypos = 50;
12.   let colorValue = 0; ←
13.
14.   for (let count=0; count<10; count++) {
15.     fill(colorValue);
16.     colorValue = colorValue + 25; ←
17.     stroke("red");
18.     strokeWeight(10);
19.
20.     circle(xpos,ypos,50);
21.     xpos = xpos + 60;
22.     ypos = ypos + 60;
23.   }
24. }
25.
```



And it is done now!

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